

Actual Course Project

Project Title

AuthSignal: AI-Based Denial Risk and Denial Reason Prediction for Prior Authorization Workflows

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1. Abstract

Prior authorization is a document-intensive healthcare administrative process in which treatment requests are reviewed before approval. Delays and denials often occur because of missing documentation, unclear supporting evidence, or failure to meet payer requirements. This project proposes AuthSignal, an AI-based system that focuses on two practical and standalone prediction modules: **Denial Risk**

Prediction and Denial Reason Classification. An optional third module, **Document Completeness Prediction**, may also be implemented if time permits.

It uses supervised learning, natural language processing, feature engineering, and explainable machine learning to build predictive models that can assist healthcare administrative staff in identifying risky cases before submission and understanding likely denial reasons. The final output will be a lightweight prototype with trained models, evaluation results, and a simple interface for prediction.

2. Introduction

Healthcare administration contains many repetitive and time-sensitive tasks. One major bottleneck is prior authorization, where providers must submit medical justification and supporting documents to insurers before treatment or procedures are approved. These requests may be denied for several reasons such as missing documentation, lack of demonstrated medical necessity, coverage limitations, or billing-related issues.

Manual review of such cases is slow and error-prone. Administrative teams must examine documents, interpret policy requirements, and decide whether a case is likely to succeed. This project addresses that problem by building AI modules that can predict denial outcomes and classify denial reasons based on structured case features and denial text.

Instead of attempting a large end-to-end system, it focuses on a smaller and achievable scope that still demonstrates core AI concepts including classification, NLP, model training, feature engineering, evaluation, and deployment.

3. Problem Statement

Prior authorization requests in healthcare are often denied due to issues such as incomplete supporting evidence, missing documents, or failure to satisfy insurer conditions. Reviewing these cases manually takes time and increases administrative burden.

There is a need for an AI-based solution that can:

- predict whether a case is likely to be denied,
- identify the likely reason for denial from text,
- optionally determine whether a document set is complete before submission.

This project aims to build standalone machine learning modules that support these tasks.

4. Objectives

4.1 Main Objective

To develop a lightweight AI prototype for prior authorization workflows that predicts denial risk and denial reason using machine learning and NLP techniques.

4.2 Specific Objectives

1. To design and prepare datasets for structured and text-based prior authorization tasks.
 2. To build a **Denial Risk Predictor** using structured features.
 3. To build a **Denial Reason Classifier** using denial letter text or short denial summaries.
 4. To evaluate models using standard classification metrics.
 5. To provide interpretable outputs for both prediction modules.
 6. To integrate the trained models into a simple interface or API.
 7. To optionally implement a **Document Completeness Predictor** if time permits.
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5. Scope of the Project

Included in Scope

- structured dataset preparation,

- text dataset preparation,
- feature engineering,
- supervised classification,
- text preprocessing,
- model training,
- model evaluation,
- explainability through feature importance and key-word influence,
- simple deployment through a basic interface or API.

Excluded from Scope

- full PDF parsing pipeline,
- OCR-heavy processing,
- named entity recognition model training,
- large language model fine-tuning,
- automatic appeal-letter generation,
- full hospital system integration,
- real insurer submission workflows.

This reduced scope ensures the project remains realistic within the available time and team capacity.

6. Proposed Modules

6.1 Module 1: Denial Risk Predictor

This is the main structured machine learning module.

Purpose

To predict whether a prior authorization request is likely to be **approved or denied**.

Input Features

Possible features include:

- procedure type,
- diagnosis category,
- age group,
- number of supporting documents,
- number of missing required conditions,
- referral attached or not,
- lab report attached or not,
- previous treatment history included or not,
- policy match score,
- note completeness score.

Output

- Approved / Denied
- denial probability score

Possible Algorithms

- Logistic Regression
 - Random Forest
 - XGBoost or Gradient Boosting
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6.2 Module 2: Denial Reason Classifier

This is the NLP-based module.

Purpose

To classify the likely reason for denial from denial letter text or short case summaries.

Possible Classes

- missing documentation,
- medical necessity not established,
- prior authorization not obtained,
- non-covered service,
- coding or billing issue,
- duplicate or miscellaneous.

Input

- denial letter text,
- short denial summary,
- case explanation text.

Output

- denial reason category
- confidence score

Possible Algorithms

- TF-IDF + Logistic Regression
 - TF-IDF + Linear SVM
 - TF-IDF + Naive Bayes
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6.3 Optional Module 3: Document Completeness Predictor

This module will only be implemented if the first two are completed successfully and time remains.

Purpose

To predict whether a prior authorization file is complete or incomplete before submission.

Input Features

- clinical note present or absent,
- diagnosis code present or absent,
- procedure code present or absent,
- lab report attached or not,
- referral attached or not,
- number of required fields missing,
- total documents attached.

Output

- Complete / Incomplete
- completeness score

Possible Algorithms

- Logistic Regression
 - Decision Tree
 - Random Forest
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7. Methodology

7.1 Data Collection

Due to privacy restrictions in healthcare, the project will use:

- synthetic case records,
- manually created structured samples,

- simulated denial letters,
- publicly inspired policy categories,
- de-identified text templates where possible.

The team will create labeled examples for both structured and text-based tasks.

7.2 Data Preparation

The following preprocessing steps will be performed:

For Structured Data

- remove inconsistencies,
- encode categorical variables,
- normalize or standardize numeric features if needed,
- generate derived features such as missing-document count and policy-condition count.

For Text Data

- tokenization,
- lowercasing,
- punctuation cleaning,
- stop-word removal where useful,
- TF-IDF vectorization.

7.3 Model Selection

Models will be selected based on simplicity, speed, interpretability, and suitability for the dataset size.

Structured Prediction

- Logistic Regression as baseline
- Random Forest as non-linear baseline

- XGBoost if time permits

Text Classification

- TF-IDF + Logistic Regression as baseline
- TF-IDF + Linear SVM for comparison

7.4 Training

The models will be trained using labeled samples and split into:

- training set,
- validation set,
- test set.

Cross-validation may be used if the dataset is small.

7.5 Evaluation

Performance will be measured using:

- Accuracy
- Precision
- Recall
- F1-score
- Confusion Matrix
- ROC-AUC for binary classification where applicable

7.6 Explainability

The project will include:

- feature importance for structured models,
 - top influential words/features for text classifiers,
 - error analysis for incorrect predictions.
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8. AI and Computing Concepts Used

This project covers the following AI and computing concepts:

Core AI Concepts

- supervised learning,
- classification,
- training and testing split,
- generalization,
- overfitting and underfitting.

Natural Language Processing

- text preprocessing,
- tokenization,
- stop-word removal,
- lemmatization,
- text vectorization,
- text classification.

Machine Learning

- Logistic Regression,
- Random Forest,
- XGBoost,
- Support Vector Machines,
- feature engineering,
- feature selection.

Evaluation and Optimization

- cross-entropy loss,
- model validation,
- hyperparameter tuning,
- cross-validation,
- confusion matrix analysis.

Explainable AI

- feature importance,
- coefficient interpretation,
- keyword influence in text models.

Backend / System Design

- API design,
 - model integration,
 - modular prediction pipeline.
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9. Tools and Technologies

Programming Language

- Python

Libraries

- pandas
- NumPy
- scikit-learn
- matplotlib
- joblib / pickle

- Flask or FastAPI
- Streamlit for interface if chosen

Development Tools

- Jupyter Notebook / VS Code
 - GitHub for version control
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12. Expected Outcomes

By the end of the project, the team expects to deliver:

1. a trained **Denial Risk Prediction** model,
2. a trained **Denial Reason Classification** model,
3. evaluation metrics and comparison results,
4. explainability outputs for both models,
5. a simple prototype interface or API,
6. a final project report and presentation.

If time allows, the team may also deliver:

7. a **Document Completeness Predictor**.
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14. Significance of the Project

This project is significant because it demonstrates how AI can assist in a real administrative healthcare problem using a practical and achievable scope. It allows the team to apply fundamental AI concepts such as supervised learning, NLP, feature engineering, prediction, and evaluation in a realistic setting. It also produces a working prototype rather than only theoretical analysis.

15. Conclusion

AuthSignal is a focused AI project designed by concentrating on two standalone modules, Denial Risk Prediction and Denial Reason Classification, the project remains manageable while still demonstrating meaningful AI techniques and predictive capability.